

AJEET KUMAR

Embedded Firmware Engineer

+91 9770193662 | ajeet.aryan198@gmail.com | <https://in.linkedin.com/in/ajeet-kumar-f95>

SKILLS:

Programming Languages	C, C++, Python
Database	MySQL, SQLite
IDE	WinCE, STM32 CubeIDE, S32 Design Studio, VS Code, Keil MDK (ARM Cortex-M)
Compilers & toolchains	GCC (arm-non-eabi-gcc), Clang, Keil C compiler
Debuggers	ST-Link, JTAG/SWD interface, Lauterbach.
RTOS	FreeRTOS
Version control	Git (with GitLab, GitHub), SVN
Project Management Tools	Jira
Documentation/Collaboration	Confluence
Simulation & Emulation	Proteus (for circuit simulation)
Communication Tools	PuTTY, TeraTerm
Hardware Description & Testing	Oscilloscopes, Multimeter
Communication Protocols	UART, I2C, SPI, USB, GPIO, BLE, Wi-Fi, PWM, ADC, DMA

WORK EXPERIENCE:

Zumi Solutions Pvt Ltd Embedded Software Developer

Jun-21 to Apr-25

- Developed scalable and high-performance drivers, middleware, compliance, SDK Integration, and Customer support.
- Developed firmware designs for embedded products by using Controllers(STM32H7 series from STM) / Processors(Toradex Colibri T20 ARM Cortex-A series), and wrote the code in C/C++.
- Developed test automation software and test scripts(in Python), performing SW optimization for memory & performance.
- Developed application-specific software demos and examples related to end equipment, which enables benchmarking custom processors and helps customers to get started on their product development.
- Able to look at software deliverables from the end user point of view – ease of use, well-documented code, and design documents. Participate in customer discussions and understand their requirements.
- Analysed customer and internal requirements and specifications and translated these into software requirements and designs.
- Worked with RTOS, bare-metal systems.
- Integrate and debug drivers, peripherals (SPI, I2C, UART, USB, BLE, etc.), and communication protocols.
- Worked with version control (GitHub, GitLab) and CI/CD pipelines for embedded development.
- Worked with Jira, a project management tool from Atlassian.
- Worked with Confluence, a documentation and collaboration tool from Atlassian.
- Troubleshoot and debug issues using JTAG, ST-Link, oscilloscopes, and logic analysers
- Developed the desktop application software with 4 team members for cross-environment in Python with Qt framework and the SQL database, pandas, and numPy.

Vector India Pvt Ltd Technical Coordinator

Dec-19 to Apr-21

- Worked as a technical coordinator, where fully filled the responsibility of bug fixing in the developed drivers, and solved the issue of the trainee engineer.

RELEVANT PROJECTS:

Zumi Solutions Pvt Ltd Ultrasonic Flaw Detection (USFD) testing Machine

Responsibilities :

- Integrated the additional channels(SRT machine from 7 to 10 channels, DRT machine from 14 to 19 channels) to read the raw data from the DPRAM and process it over the Toradex Colibri T20 processor to identify the track defect on the A-scan and B-scan screens, written in C/C++.
- Integrated the GPS, USB, and Serial thread for communication with hardware.
- Integrated the odometer to read the distance by using the Arduino board.
- Fixed the memory issue where the recorded data takes up a lot of space.
- Increased the speed of data processing by changing the data processing algorithms(from 32-bit to 16-bit memory).

Desktop Application

Responsibilities :

- Developed a desktop cross-environment application in Python with the Qt framework to identify the recorded binary data from the Ultrasonic Flaw Detection(USFD) testing machine.
- Used the SQL database, pandas, and numPy to arrange and store the data in a proper arrangement.

Ophthalmic Phacoemulsifier Machine

Responsibilities :

- Developed the Ophthalmic Phacoemulsifier(RTOS-based system) by using the STM32H7 series microcontroller written in C/C++.
- Integrated the motor driver using by I2C protocol.
- Controlled the motor driver by using potentiometer data with ADC(from DMA) and a PWM interface.
- Integrated the LCD touch panel with UART configuration.
- Integrated the machine operations with an Android application through Bluetooth and Wi-Fi.
- Integrated the orbital handpiece, multifunction footswitch, anterior vitrectomy, etc.

Fault Detection Isolation and Recovery(FDIR)

Responsibilities :

- Generate the signals to control and rectify the drivers for the module on the pre-build system code(based on STM32H7 series microcontrollers from STM32 and written in C++) and test it on the COSMOS interface.
- Test the written scripts and write the allotted scripts from the customer, and test these scripts on the COSMOS interface.

Vector India Pvt Ltd Techniquial Coordinator

- Worked as a technical coordinator, where fully filled the responsibility of bug fixing in the developed drivers, and solved the issue of the trainee engineer.
- Built some industry-level projects like weather monitoring, RFID-based toll plaza, student database based on the RFID.

EDUCATION & CERTIFICATION:

Bachelor of Engineering - Electronic and Communication Engineering

2012 - 2017 | Rajiv Gandhi Proudlyogiki University, Bhopal, MP

Training & Certification on Embedded System Engineer

15 March 2019 - 30 November 2019 | Vector India Pvt Ltd, Bengaluru, Karnataka